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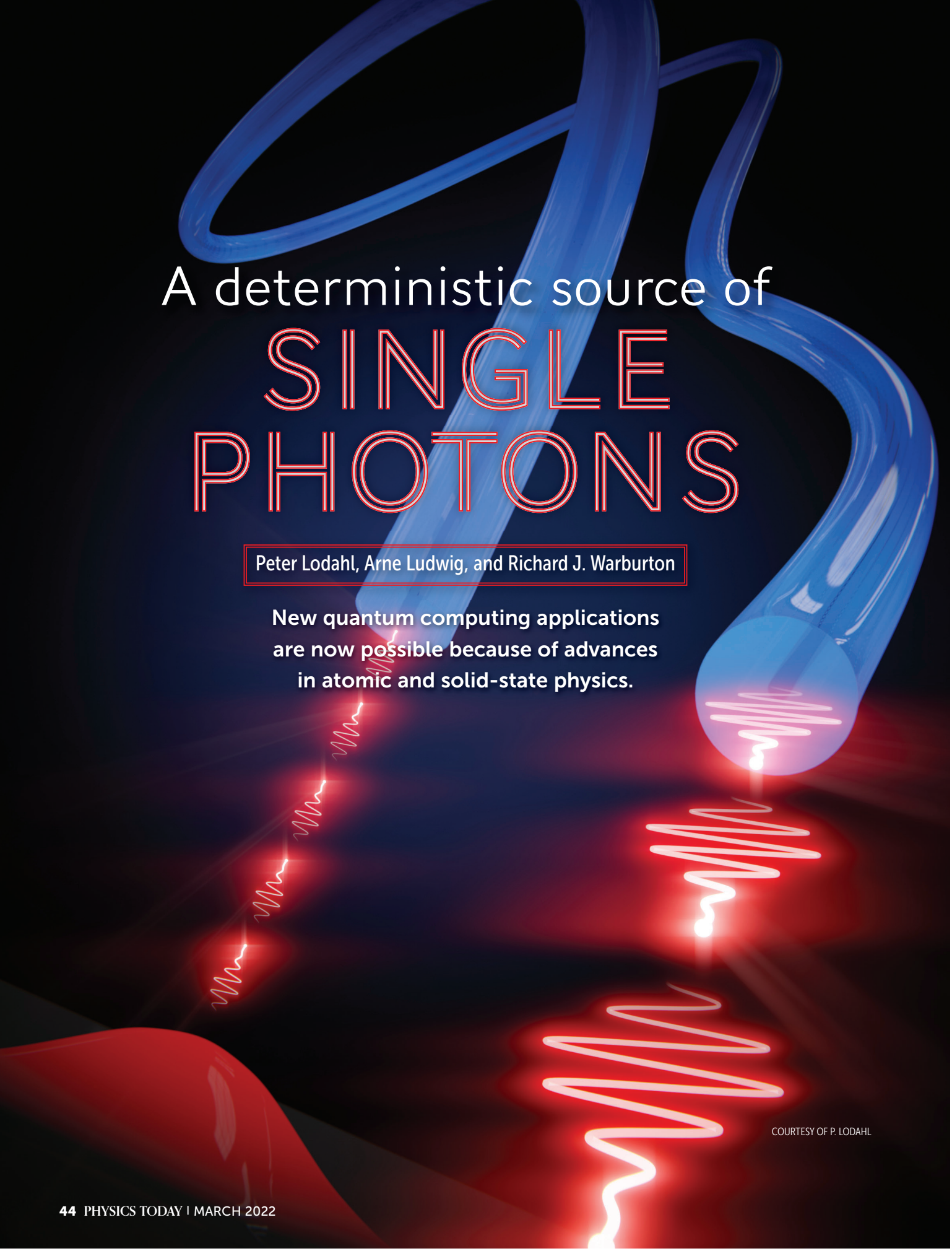
A deterministic source of SINGLE PHOTONS

Peter Lodahl, Arne Ludwig, and Richard J. Warburton

New quantum computing applications are now possible because of advances in atomic and solid-state physics.



COURTESY OF P. LODAHL

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Photons are the quantum constituents of light and are fundamental to the quantum theory of light and to the light–matter interaction. The quantum information stored in photonic qubits can be sent over large distances using photons at optical frequencies. But photons are elusive: They travel at the speed of light, are often created by spontaneous emission, are susceptible to propagation loss, and do not interact with each other. Although creating and controlling single photons is challenging, the benefits to quantum information applications are significant.

The merger of atomic and solid-state physics has led to new opportunities in quantum photonics, specifically in developing a deterministic source of single photons. Advances rely on state-of-the-art growth of semiconductor heterostructures, nanofabrication, and optical techniques. One fundamental improvement that offers a scalable route to advanced quantum applications is a coherent photon-emitter interface. The technology enables multiphoton entanglement generation and deterministic photon–photon quantum gates.

The granularity of light and matter, proposed more than a century ago by Max Planck, Albert Einstein, and Niels Bohr, lies at the core of quantum mechanics. The quantized nature of the electromagnetic field extends James Clerk Maxwell's classical description. The quantum particle of light is referred to as the photon and constitutes the fundamental entity by which light and matter exchange energy. Bohr's 1913 atom model describes the birth of a photon: A single atom may hop from an excited electronic state to a lower one. That process of spontaneous emission creates a photon.

Much later, Edward Purcell realized that spontaneous emission is not an immutable property of the atom. Rather, one can control it by engineering the atom's environment. Purcell's profound idea underpins many opportunities in photonics, including the operation of single-photon sources. When an atom is embedded in a tailored waveguide or cavity, the generated photons are funneled rapidly and with near-unity efficiency into a single optical mode. At birth, the photon is sent in a predefined direction. Implementing photons in that way requires control of the atom's environment at optical frequencies that correspond to nanometer-scale wavelengths. Today, deterministic single-photon sources, which are available in a number of research labs, generate photonic qubits on demand for photonic quantum information processing.

Artificial 1D atom

An ideal single-photon source creates a photon deterministically when triggered by a laser or electrical pulse. The workhorse in quantum optics has been the spontaneous parametric down-conversion source in which the energy of a photon from a laser is used to create two separate photons. The source is simple to operate. It requires only a pulsed laser and readily available nonlinear crystals, and the detection of one photon heralds the creation of the other. The source, however, has a key drawback: It is inherently probabilistic, meaning that the photons cannot be produced on demand.

An alternative approach creates single photons with an atom. In the simplest case of an atom with two energy levels, a photon is produced each time the atom decays from the upper energy state to the lower state. In free space, photons are emitted in all directions or into a continuum of optical modes. A useful source, however, creates photons in just one optical mode. To develop such a source, two approaches can be pursued, based on either cavities or waveguides. For tightly confined modes, the atom decays preferentially into a single mode in the cavity or waveguide. The collected photon can be subsequently coupled into a single-mode optical fiber.

One implementation of those approaches uses an artificial atom, which takes the form of a semiconductor quantum dot.[1,2] The quantum dot is grown by self-assembly using the III–V semiconductors indium arsenide and gallium arsenide. When the low bandgap, high lattice-constant InAs is grown on top of the high bandgap, low lattice-constant GaAs, the lattice mismatch induces strain. The strain leads to the self-assembly of an InAs island, the quantum dot shown in figure 1. Quantum dots are typically 20 nm in diameter at the base and 5–10 nm high with a potentially complex topology.

In a semiconductor, the bandgap separates occupied continuum valence states from unoccupied continuum conduction states. A quantum dot confines valence and conduction electrons to a narrow spatial region, shown in figure 2. As a result, discrete energy levels develop. The wavefunctions of the energy levels have a spatial extent that's determined by the size

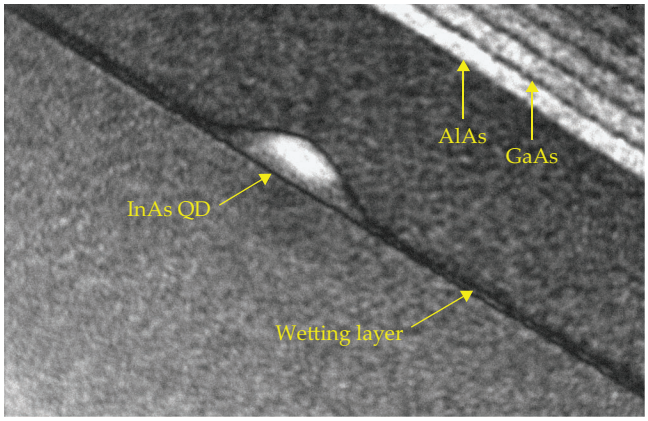

FIGURE 1. THIS IMAGE, taken with a transmission electron microscope, shows an indium arsenide quantum dot (QD) in gallium arsenide. (Courtesy of Jean-Michel Chauveau and Arne Ludwig.)

of the quantum dot. A photon can promote a valence electron to the conduction level, which leaves a vacancy, or hole, in the valence level, and the resulting electron–hole pair is termed an exciton. The ground state and the exciton constitute a two-level system. Because the exciton's optical dipole moment is related to the quantum-dot size, it is much larger than the size of a single atom. That is advantageous because the radiative lifetime of a quantum-dot exciton is rather short, typically a nanosecond.

A self-assembled quantum dot is trapped inside its host semiconductor, GaAs, which is a huge benefit. Because the quantum dot is locked at one particular location, a laser trap is not required, as is the case for single atoms or ions in a vacuum. The semiconductor environment, however, is a potentially complex source of noise, as figure 2 illustrates. The evolution of an exciton is disrupted by the thermal wobbling of the atoms, known as phonon scattering, in the quantum dot and by charge and spin noise in the host semiconductor. Charge and spin noise typically have correlation times much longer than the radiative lifetime, which leads to variability of the exciton's frequency. In contrast, phonon scattering randomly dephases the exciton before recombination.

A crucial feature of the GaAs system is that the complex and deleterious noise processes can be ameliorated. Specially designed heterostructures[3] reduce charge noise to extremely small levels at low temperature, and the GaAs system's performance is retained even in nanostructures.[4] Likewise, phonon scattering is suppressed at low temperature, but it is not completely eliminated. The creation of an exciton locally distorts the semiconductor lattice, which means that some phonon scattering remains even at absolute zero, although it tends to be slow relative to the radiative decay time. The net result is that the exciton mimics a two-level system. When resonantly driven with a laser, single quantum dots exhibit all the features known from atomic physics, such as photon antibunching, Rabi oscillations, and the Mollow triplet.

Single InAs quantum dots in GaAs are the semiconductor workhorses of two-level systems. Decades of work on the quantum Hall effect in two-dimensional electron gases has led to the creation of extremely clean GaAs-based heterostructures. The same technology has been applied to low-noise quantum-dot devices for quantum photonics. Typically, the quantum dots emit radiation at wavelengths between 900 nm and 1200 nm. Unlike single atoms, each quantum dot emits at a slightly different wavelength. Nevertheless, tuning techniques exist, and the ultimate goal is to tune most of the quantum dots in a chip to a common wavelength. Research is ongoing to solve the long-standing problem of quantum dots nucleating at random locations. Rapid progress has been made in creating low-noise quantum dots at other wavelengths, notably at red wavelengths (780 nm) and at those relevant for telecommunications (1300 nm and 1550 nm). Whereas the latter targets low-loss propagation in optical fibers, the former can be coupled to atomic rubidium memory cells.

The creation of cavities and waveguides exploits a special feature of GaAs: A partner material, aluminum arsenide, has almost the same lattice constant but different electronic and chemical properties. The significantly lower refractive index of AlAs enables a Bragg mirror to be created via a stack of layers, each of which is one-quarter of a wavelength thick. Then a quantum dot can be embedded between two such mirrors to confine the light field along the growth direction. Lateral confinement can be realized by etching a miniature pillar, a so-called micropillar.[5,6] Alternatively, figure 3 shows how a miniaturized dielectric mirror is used as the top mirror.[7,8]

A waveguide can be created by growing an AlAs layer below the active part of the GaAs heterostructure. The subsequent chemical removal of the AlAs yields a free-standing GaAs membrane, shown in figure 4. The high refractive index of GaAs results in laterally propagating modes confined to the membrane. A photonic crystal lattice can contain photonic bandgaps where no optical modes are allowed, and a thin unstructured

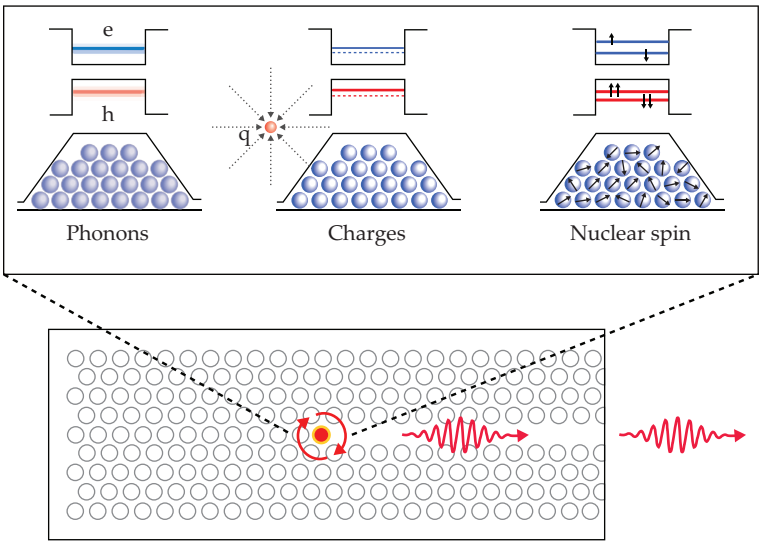

FIGURE. 2. THIS SCHEMATIC shows the bound electron e and hole h states in a quantum-dot exciton and their responses to noise processes. Phonons induce thermal fluctuations of the atomic lattice, residual charges cause electrostatic fluctuations, and nuclear-spin noise arises from spin–spin coupling between the exciton and the randomly polarized nuclear-spin ensemble of the quantum dot. The quantum dot (red circle in the bottom figure) is embedded in a photonic crystal waveguide membrane and emits single photons (red wavepackets). (Adapted from ref. 18.)

region can constitute a waveguide,[9] in which the dispersion of light is engineered by the photonic-crystal structure.

Figures of merit

Advanced single-photon applications require high-performance sources. Ideally, all of a source's figures of merit should have high values simultaneously, which is realizable with quantum-dot single-photon sources.[2,10]

Single-photon purity quantifies to what extent an emitted pulse contains only one photon. The absence of any two-photon coincidence event signifies an ideal single-photon source. Greater than 99% purity is typically obtained with quantum-dot sources that are limited by a small probability of two-photon emission. Single-photon purity may be further improved with optimized excitation schemes.

Photon indistinguishability quantifies to what extent the individual photons in a photon stream are identical. Two identical photons can interfere perfectly on a beamsplitter, leading to vanishing coincidence events. Residual coincidences quantify the indistinguishability through the interference visibility V. A single quantum emitter can produce a massive photonic resource: Near-unity indistinguishability[11] has been achieved with quantum-dot sources that extend over long strings of more than 100 photons.[8,12] Additionally, $V = 93\%$ has been achieved on interfering photons from two separate quantum dots.[13] The results demonstrate how quantum-dot sources generate low noise over a wide-frequency bandwidth.

Single-photon generation rate specifies the number of photons that can be created per second. The operation speed of the sources is ultimately limited by the radiative lifetime of the emitter, which reaches the 20–100 picosecond range in Purcell-enhanced cavities and waveguides.[2,7] Quantum-dot sources, therefore, can operate at a repetition rate exceeding 1 GHz.

Photon-emitter coupling is quantified via the β factor. It expresses the probability that an excited quantum dot emits a photon into the designated mode. The β factor depends on the success with which one can tailor the quantum-dot environment. Values of 96–99% have been realized with quantum dots in nanophotonic waveguides[2,12] and cavities.[7,10]

Out-coupling efficiency assesses the effectiveness of extracting photons from the device. The relevant parameter depends on the application, but one key quantity is the coupling efficiency from the device into an optical fiber. An efficiency of 57% has been realized by combining a high β value and good mode matching to the fiber.[8]

Growing high-quality quantum dots

Figure 5 shows modern semiconductor devices created with layer-by-layer epitaxy of thin films on ultraclean, single-crystal substrates. The manufacturing process allows for the creation of heterostructures in which layers of dissimilar materials are stacked on top of each other. The combination of materials with different bandgaps and electrical doping forms devices with new functionalities.

One high-precision and ultraclean method to produce such semiconductor heterostructures is molecular-beam epitaxy (MBE). In that method, evaporation sources filled with ultrapure elemental charges—purified up to 99.999999% in the case of gallium—are used to create atomic beams. The atoms adsorb to a crystal substrate, and the resulting adatoms form layers of near-perfect crystalline arrangement. To avoid contamination, the process takes place in an ultrahigh vacuum with a pressure 10^{-14} that of ambient air. Fewer molecules are found in such extreme conditions than, for example, in the vacuum of space around the International Space Station. The cleanest crystals made by MBE have impurity concentrations of about 0.1 ppb. The relevant dimensions for a quantum dot and its immediate environment are on a 100 nm length scale, so the active parts of the devices are essentially free of impurities.

The substrates are heated to enable the adatoms to move freely over the crystal surface such that the growth of one monolayer is completed before the growth of the next begins. Several techniques improve the crystal quality, including short growth interruptions and temperature regimes that prevent the growth of specific species. With the tremendously high material purity and the control of layer thicknesses and arrangements down to the atomic level, MBE is a critical enabler of modern nanotechnology.[14] Although the method allows for the creation of ultrahigh reflectivity Bragg mirrors and thin GaAs membranes, another operating principle is required to form quantum dots—self-assembly of 3D nanostructures.

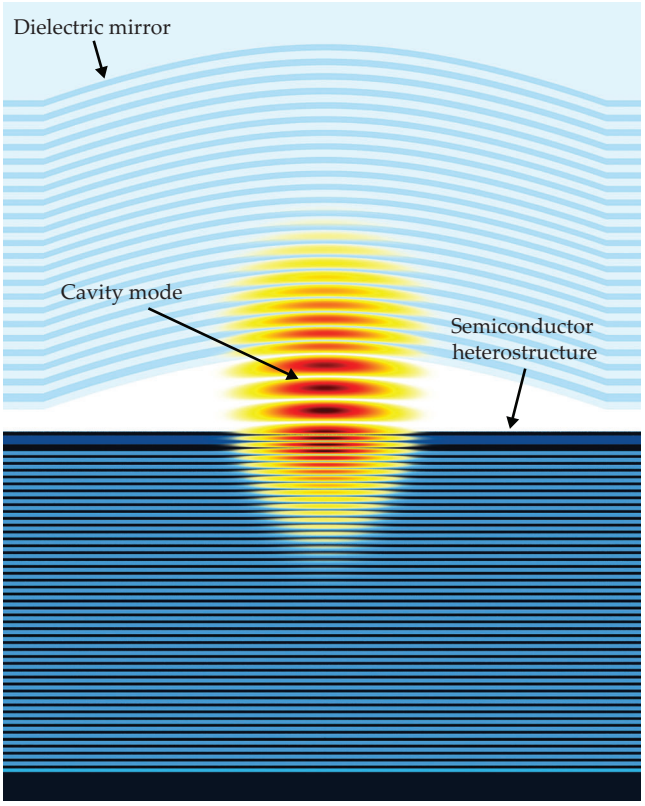

FIGURE 3. THIS VERTICAL-CAVITY DEVICE is an illustration of a semiconductor heterostructure that consists of a gallium arsenide and aluminum arsenide Bragg mirror (bottom) and a p-i-n diode. The InAs quantum dots are located in the intrinsic i region in tunnel contact with the Fermi sea in the n-layer. The concave dielectric mirror (top) is micro-machined in a silica substrate. A single quantum dot is located at the exact center of the cavity mode. The position of the heterostructure can be adjusted with respect to the top mirror by using an *x-y-z* nanopositioner. It ensures that the quantum dot is centered and that the frequency of the quantum-dot exciton matches that of the cavity's fundamental mode. (Adapted from ref. 7.)

To stack layers of dissimilar materials on top of each other, several parameters must conform with one another. One is the lattice constant, the size of the crystal's unit cell. If InAs, a material with a relatively large lattice constant, is stacked on top of GaAs, a material with a smaller lattice constant, elastic strain builds up. After a certain amount of accumulated strain, instead of continued layer-by-layer growth, the surface breaks up. Dome-shaped indium-rich islands form that each contain about 100 000 atoms. The islands, or quantum dots, nucleate at random positions on the GaAs substrate following the deposition of 1.5 monolayers of InAs.[15]

Another method to form quantum dots via self-assembly is to create nanometer-sized metallic droplets on an alloy such as AlGaAs. The droplets can be recrystallized or used alternatively to drill tiny holes in the surface. The holes are subsequently filled with GaAs, resulting in inverted domes. The quantum dots consist of GaAs in an AlGaAs matrix and emit photons with a higher energy than InAs quantum dots.

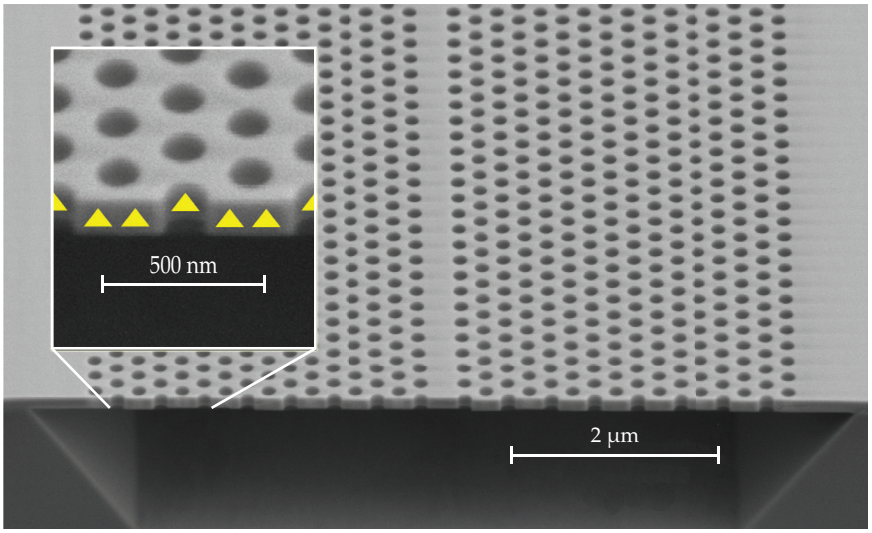

FIGURE 4. THIS ON-CHIP WAVEGUIDE, taken with a scanning electron microscope, consists of a hexagonal arrangement of holes etched in a gallium arsenide membrane. The row of missing holes constitutes the waveguide. The yellow triangles (inset) indicate the position of the quantum dots in the center of the membrane. (Adapted from ref. 2.)

Fluctuating charges in the vicinity of the quantum dot lead to noise. They vary the electric field around the quantum dot, which in turn results in variations in photon energy. Even worse are fluctuations in the charge of the quantum dot itself. If one captures a single electron, the photon energy is strongly redshifted such that a resonantly driven quantum dot is no longer excited, and the single-photon source shuts off. The source turns back on only when the extra electron is released. Under that scenario, the photon stream contains telegraph noise, or blinking.

To minimize charge noise, the host material must be as clean as possible. One elegant way to stabilize the quantum dot's charge exploits the Coulomb blockade.[1] A quantum dot in close proximity to a Fermi sea can be controlled by a bias field. At low temperature, the singly charged quantum-dot state lies above the Fermi energy and is therefore unoccupied. To realize such a structure requires a layer made of doped GaAs or AlGaAs. Provided that the doping level is high enough, a Fermi sea forms at low temperature when every 10 000th crystal-matrix atom is replaced by an impurity atom. That amount is so low that the crystal remains in a perfect arrangement and stays highly transparent to single photons. Silicon is an excellent choice for electron doping (n-type); and carbon, for hole doping (p-type). Both are used in n-i-p devices in which the grounded n-layer hosts the Fermi sea, the quantum dots are located in the intrinsic i-type layer, and a bias is applied to the p-layer.[1]

Vertical-cavity structures

Once quantum dots are grown, the next step is to make an efficient source of single photons. A high β factor can be achieved in a resonant cavity. The requirements are a small cavity-mode volume of order λ^3, where λ is the photon's free-space wavelength, and a reasonably long photon lifetime.

The model system is described with the Jaynes–Cummings Hamiltonian. It consists of a two-level system, a single cavity mode, a coherent coupling rate g, and two decay processes: unwanted emission of the atom into noncavity modes (rate γ) and leakage out of the cavity (rate κ). A quantum-dot cavity system can be brought into the strong-coupling regime where $g \gg \gamma$ and $g \gg \kappa$. The cooperativity, $C = 2g^2/(\kappa\gamma)$, is a measure of coherent coupling efficiency. An ultrahigh C of 150 has been achieved,[7,8] which is one of the highest cooperativities reached with a single emitter at optical frequencies. That regime is potentially useful for photon–photon gates. A single-photon source, however, works better in the weak-coupling regime ($\kappa > g \gg \gamma$) that exploits the large β factor of $\beta = 2C/(1 + 2C)$. If κ is dominated by leakage through the top mirror, the conversion efficiency of a quantum-dot exciton to a photon exiting the cavity is maximized by choosing $\kappa = 2g$.

The weak-coupling regime has been implemented with semiconductor micropillars.[5,6,10] End-to-end efficiency, however, is currently highest (> 50%) with an open microcavity,[8] shown in figure 3. That device has a photon indistinguishability of 97.5% and a purity of 98%. The strong Purcell effect results in a radiative lifetime of just 50 ps, which allows a photon to be created each nanosecond.

The open-microcavity design has enabled researchers to optimize many parameters simultaneously. First, the design's tunability allows a quantum dot to be brought into exact spectral and spatial resonances with the cavity mode. That capability addresses the weakness of the self-assembly process's lack of control in the exact emission frequency and spatial position. Second, the cavity losses are dominated by those through the top mirror. Third, the design is compatible with an n-i-p structure. The charge noise is extremely low during operation, and the quantum-dot charge is locked by Coulomb blockade. Lastly, the output mode is a simple Gaussian and is therefore naturally matched to the propagating mode in the output fiber. The open-microcavity device showcases what quantum dots

can achieve: fast and bright creation of high-quality single photons at the output of a standard single-mode fiber.

Planar nanophotonic waveguides

Vertical-cavity structures are necessarily narrowband. Only quantum dots at the cavity resonance of a few gigahertz linewidth emit photons deterministically into the cavity mode. The others emit into noncavity "leaky" modes. Planar nanophotonic devices work in an orthogonal way: Emission in the vertical direction is suppressed, and emission in a single propagating lateral mode is encouraged.[9] The lateral mode is part of a 1D continuum of broadband operation. That approach creates single photons in a specific mode in the chip and offers a pathway to single-photon sources integrated on a chip. Ultimately, fully integrated quantum processors may be possible in which the sources are combined on the chip with advanced processing circuits and high-efficiency detectors.

The planar platform is based on photonic membranes with a thickness of less than half the targeted optical wavelength, as shown in figure 2. Light is confined to the lateral plane, and the refractive index contrast between the membrane material and the surrounding vacuum strongly suppresses out-of-plane light leakage and emission by the process of total internal reflection. A 2D photonic-crystal lattice controls the in-plane light emission by photonic bandgap effects. The coupling into the waveguide mode is Purcell-enhanced by the dispersion-engineered waveguide mode that features slow light. Values of β greater than 98% are possible because of the suppression of leaky modes and Purcell enhancement of the waveguide.[2]

Fabricating thin-membrane structures and photonic crystals for ultralow-noise devices is challenging. Figure 4 shows representative physical dimensions: A 150-nm-thin GaAs membrane contains quantum dots in the center, and a photonic-crystal lattice of etched holes has a lattice parameter of 260 nm. With those dimensions, the embedded quantum dots are positioned unavoidably close to free surfaces, which could potentially cause fluctuations via uncontrolled electronic surface states.

Those challenges have been overcome in nanophotonic membranes made from an electrically contacted n-i-p device. Despite the extreme miniaturization, the exquisite control of the doping and postgrowth fabrication reduces the quantity of current leaking from the devices to the level of nanoamps. A highly sensitive measure of the total noise is the optical linewidth of the quantum dots. It is responsive to broadening processes over a wide range of time scales covering subnanosecond (phonon scattering), microsecond (spin noise), and millisecond (charge noise). Emission lines with less than 15% residual broadening beyond intrinsic spontaneous-emission broadening have been reported on quantum dots in photonic-crystal waveguides.[4] Other accomplishments include long strings of more than 100 indistinguishable photons with no signature of coherence degradation, V greater than 96%, and highly efficient chip-to-fiber outcoupling techniques.[12]

Advanced photonic applications

Photonic hardware with more advanced functionality than on-demand, single-photon generation benefits from the elaborate device engineering implemented for single-photon sources. Coherently controlling a single spin in the quantum dot[1] has led to new opportunities. Spin-photon entanglement, for example, can be realized by performing spin operations between excitation and emission. If the process is repeated multiple times, a multiphoton entangled state is created. Different entanglement structures can be realized, including Greenberger-Horne-Zeilinger states and 1D photonic cluster states.[16,17]

The approach may be extended to generate higher-dimensional photonic cluster states for measurement-based quantum computing. Multiphoton entanglement sources require a coherent spin that can be manipulated with high fidelity. Whereas single spins in self-assembled quantum dots have relatively short coherence times, typically of a few microseconds, rapid spin control with short optical pulses means that high-fidelity multiphoton entanglement is within reach.[18]

Another opportunity exploits a single spin in a quantum dot as a photonic quantum gate. The spin represents a quantum memory whereby two successively emitted photons can become entangled. Such a photon–photon quantum gate has been a missing component in quantum photonics, and the nonlinearity of the photon-emitter coupling enables it. Ultimately, a fully deterministic photon–photon quantum gate requires researchers to pursue the challenging task of eliminating all unwanted losses. A heralding approach may relax such requirements. In that method, the gate operation is conditioned on the detection of a photon.

High-quality single-photon qubits are critical in the burgeoning area of quantum technology.[18] For quantum communication, single photons are the natural carriers of quantum information over long distances. For other

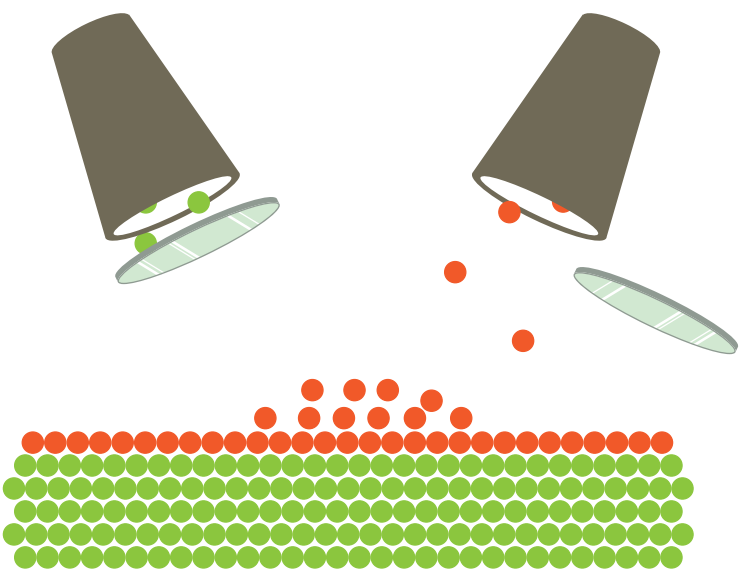

FIGURE 5. QUANTUM DOTS can be self-assembled using molecular-beam epitaxy. A stream of gallium atoms (green dots) and indium atoms (red dots) is created in two heated cells (brown cones). The ultrahigh vacuum environment is arsenic rich. By opening and closing the cell shutters, the growth is switched from GaAs to InAs. GaAs grows layer by layer. In contrast, the deposition of InAs on GaAs results in the formation of nanoscale islands, or quantum dots, each of which are connected by a thin InAs wetting layer. (Courtesy of Arne Ludwig.)

applications, such as trusted-node quantum-key distribution, coherent single photons are not strictly required. But they do offer a route to ultimately secure quantum cryptography, where security against any hacking attacks is confirmed by the violation of a Bell inequality. (See the article by Marcos Curty, Koji Azuma, and Hoi-Kwong Lo, PHYSICS TODAY, March 2021, page 36.)

Another line of research concerns loss-robust encoding of photonic qubits for quantum communication. The idea is that a single qubit can be encoded nonlocally in a multiphoton cluster state. The encoding redundancy means that a qubit is more resilient to photon loss and can therefore be sent over extended distances. Such encoding is a precursor of a one-way quantum repeater, which allows quantum information to be transmitted faithfully over any distance. Such a device would form the backbone technology of a quantum internet.

Measurement-based quantum computing architectures appear well suited for photonics. The overall challenge is to create large-scale multiphoton entanglement that is subsequently consumed during computation. Importantly, only single-qubit operations on the entangled state are required, which circumvents the need for direct photon–photon interactions. Quantum-dot deterministic sources may be exploited as a highly resource-efficient way of producing multiphoton entanglement, an attractive alternative to probabilistic spontaneous parametric down-conversion sources that require massive multiplexing capabilities.

An optimal strategy may be to use a single quantum-dot source to create small-scale entangled cluster states that are demultiplexed from the overall string of photons produced by the source. The clusters could subsequently be fused together in linear-optics photonic circuits to grow a universal resource state for photonic quantum computing. In the fusion-based quantum computing paradigm, computation proceeds by measuring the photons constituting the entangled states. Photons are consequently consumed during computation. Then the highly loss-tolerant encoding schemes, which are an essential trait of a photonic quantum computing architecture, can be implemented.

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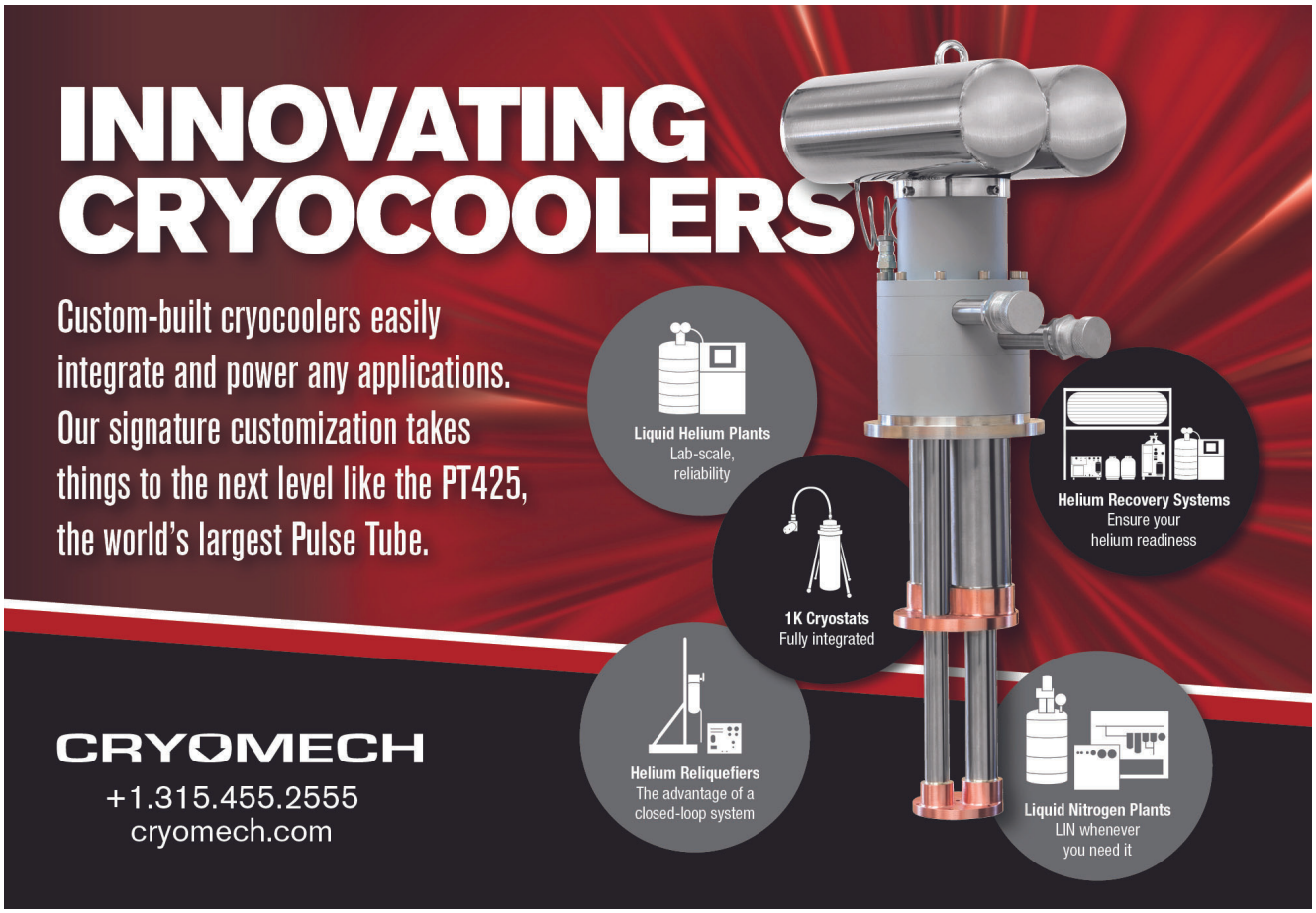
REFERENCES

1. R. J. Warburton, *Nat. Mater.* **12**, 483 (2013).
2. P. Lodahl, S. Mahmoodian, S. Stobbe, *Rev. Mod. Phys.* **87**, 347 (2015).
3. A. V. Kuhlmann et al., *Nat. Phys.* **9**, 570 (2013).
4. F. T. Pedersen et al., *ACS Photonics* **7**, 2343 (2020).
5. C. Santori et al., *Nature* **419**, 594 (2002).
6. A. Dousse et al., *Nature* **466**, 217 (2010).
7. D. Najer et al., *Nature* **575**, 622 (2019).
8. N. Tomm et al., *Nat. Nanotechnol.* **16**, 399 (2021).
9. T. Lund-Hansen et al., *Phys. Rev. Lett.* **101**, 113903 (2008).
10. P. Senellart, G. Solomon, A. White, *Nat. Nanotechnol.* **12**, 1026 (2017).
11. X. Ding et al., *Phys. Rev. Lett.* **116**, 020401 (2016).
12. R. Uppu et al., *Sci. Adv.* **6**, eabc8268 (2020).
13. L. Zhai et al., https://arxiv.org/abs/2106.03871.
14. W. P. McCray, *Nat. Nanotechnol.* **2**, 259 (2007).
15. D. Leonard, K. Pond, P. M. Petroff, *Phys. Rev. B* **50**, 11687 (1994).
16. I. Schwartz et al., *Science* **354**, 434 (2016).
17. N. H. Lindner, T. Rudolph, *Phys. Rev. Lett.* **103**, 113602 (2009).
18. R. Uppu et al., *Nat. Nanotechnol.* **16**, 1308 (2021).

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